# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

# PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION - A Multiple Choice Questions

> Note: Q. 1 Choose the correct answer for each from the given options.

Q1. The magnitude of force between two unit charges when the distance between them is 1 meter would be:

(a) 0.N
(b) 1 N
(c) 2 N
(d) Coulmb's Constant

Q2. If the area of the parallel plate capacitor is doubled then the capacitance will be:

(a) Remain uncharged
(b) half
(c) Double
(d) Four times

Q3. In an Electric circuit when Electrons move from low to high potential they will.

(a) Gain Energy
(b) Lose their identity
(c) Lose Energy
(d) Gain potential

Q4. Oscillation are damped due to the presence of:

(a) Linear motion
(b) Restoring force
(c) Frictional Force
(d) Mechanical Force

Q5. In an oscillating pendulum, the bob accelerates from its extreme position due to:

(a) Inertia
(b) Tension in the string
(c) Wind
(d) Gravitational force

Q6. Which type of wave can not travel through a Vacuum

(a) Sound Waves
(b) Infra-red radiation
(c) Microwaves
(d) X-rays

Q7. Sound is a form of:

(a) Electrical energy
(b) Mechanical energy
(c) Thermal energy
(d) chemical energy

Q8. Ohm's law is not applicable to:

(a) Semi-conductors
(b) D.0 circuit
(c) Small resistors
(d) High current

Q9. The direction of induced e.m.f in a circuit is accordance with conservation of:

(a) Mass
(b) charge
(c) momentum
(d) energy

Q10. The splitting of a heavy nucleus into smaller nuclei is called:

(a) Fusion
(b) Fission
(c) Half-life
(d) Gamma-decay

Q11. The type of electromagnetic wave used in security scanners at night is:

(a) Infra-red
(b) Microwaves
(c) Radio waves
(d) X-rays

Q12. The color that is least deviated by a prism.

(a) Violet ray
(b) Green ray
(c) Red ray
(d) Yellow ray

## Section-B (Short Answers)

> Note: Answer any EIGHT of the following questions. Each question carries 03 marks.

Q2. Define the term transverse wave.

Q3. Derive the formula of Wave speed, V =

Q4. How is the sound produced?

Q5. Define, Infrasonic and Ultra Sound.

Q6. Calculate the speed of sound in air at 50 degrees C? Given that speed of sound at 0 degrees C is 331m sec.

Q7. Define dispersion of light

Q8. What are electromagnetic waves?

Q9. Why is a normal eye not able to see the object put closer than 2.5cm.

Q10. What is the electric force of repulsion between two electrons at a distance of 1m?

Q11. Determine the function of transformers in the process of moving electrical current from the power plant to your home.

Q12. What do you understand by the half life of a radioactive element.

Q13. Describe the structure of an atom

## SECTION - C Descriptive Answers

> Note: Answer any FOUR of the following questions. Each question carries 06 marks.

Q14. List the physical properties of different isotopes of an element that are different.

Q15. What is Internet? And also differentiate between Data and Information,

Q16. A wire carrying 4A current and has length of 15cm between the poles of a magnet is kept at an angle of 30 degrees to the uniform field of 0.8 T. Find the force acting on the wire.

Q17. Explain how an electroscope is built and how it operates.

Q18. What do you understand by the term total internal reflection?

Q19. Write notes on any TWO of the following: (a) Electrostatic Potential (b) Parallel Combination Circuit (c) SimplHarrnonlc motion